

WASP-135b

R-0179

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_250702 · created 2026-07-16T18:59:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460858.903 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.05
Transit depth (Rp/Rs)^2	2.73 +/- 1.65 [%]
Orbital Inclination (inc)	81.03 +/- 2.96
Airmass coefficient 1 (a1)	1.028 +/- 0.1
Airmass coefficient 2 (a2)	-0.016 +/- 0.031
Scatter in the residuals of the lightcurve fit is	9.84 %
Best Comparison Star	#3 - [433, 370]
Optimal Aperture	2.76
Optimal Annulus	11.27
Transit Duration (day)	0.049 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.165 $\pm$ 0.05 vs 0.139 (deviation 0.3 σ)
Duration fit vs published	0.049 d vs 0.0687 d (deviation 0.45 σ)
Mid-time O-C	-35.4 min
Depth SNR	1.9
Residual scatter	9.84 %

Light curve

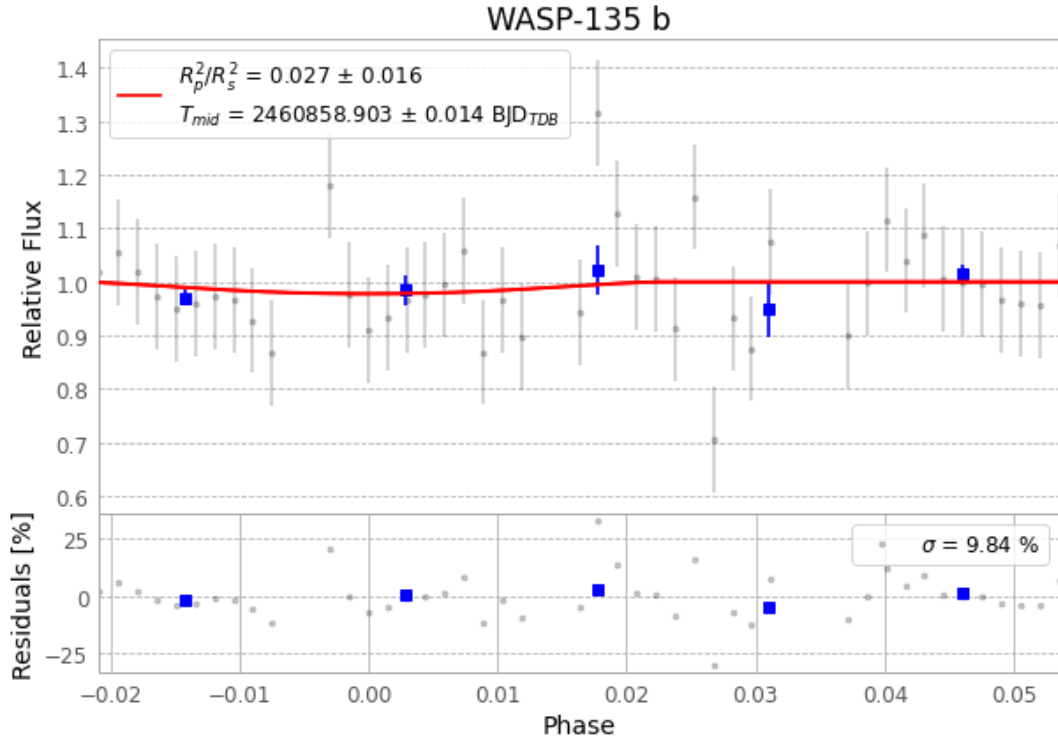


Figure 1 — FinalLightCurve_WASP-135 b_2025-07-02.png (SHA-256 24e2357a0b22298e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 302], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 02618970f10b5b9f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ee7e3ca

Data provenance

51 FITS frames (33 MB), WASP-135250702085115.FITS → WASP-135250702112116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250702.log (1620659f9a8b...), FinalLightCurve_WASP-135 b_2025-07-02.png (24e2357a0b22...), FinalLightCurve_WASP-135 b_2025-07-02.csv (0fe47d52b9e3...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 18:59Z.